

SLAM EX 2 - Ayala Houri, Nir Ellor Waizner

[Github](#)

2.1.1

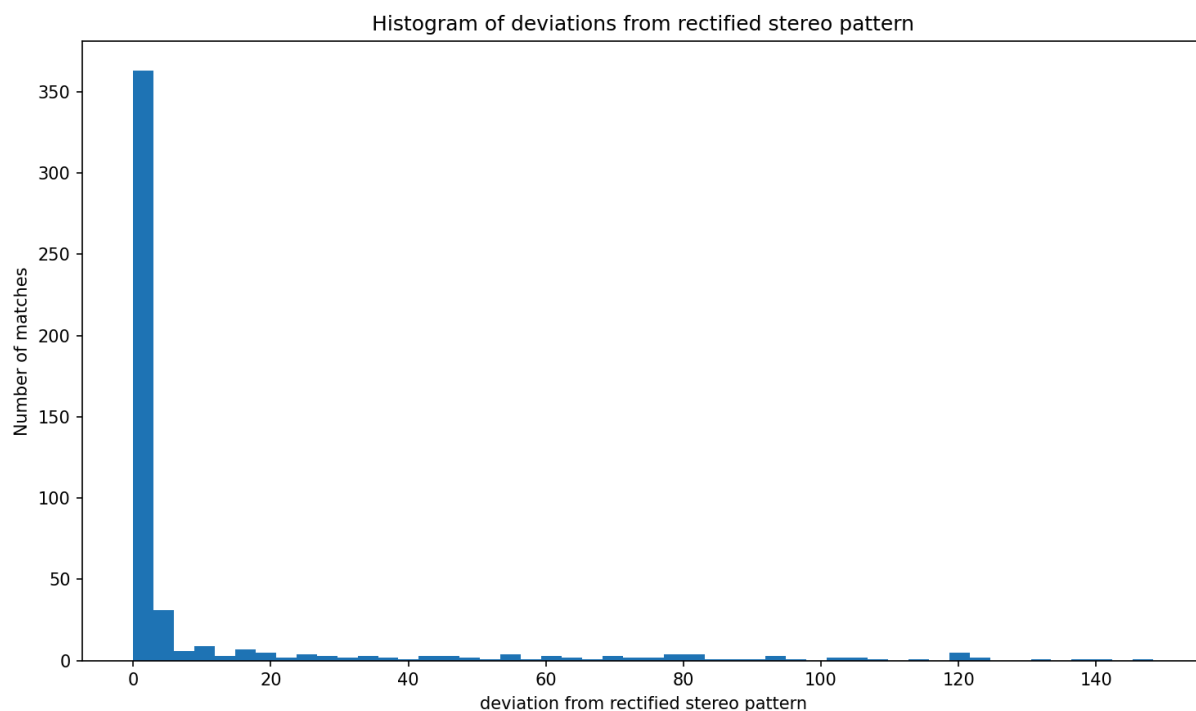
In rectified stereo images, correct matches follow a specific pattern: corresponding points in the left and right images lie on the same horizontal line, meaning their vertical coordinates are approximately equal $y_{\text{left}} \approx y_{\text{right}}$.

This pattern arises from epipolar geometry. For any point in one image, its corresponding point in the other image must lie on an epipolar line, which represents all possible locations of the matching point. This line is formed by the intersection of the epipolar plane (defined by the two camera centers and the 3D point) with the image plane.

The rectification process aligns the epipolar lines to be horizontal and identical in both images. As a result, the correspondence search is reduced from two dimensions to one dimension (along the x-axis), simplifying the matching process.

In practice, small deviations from this pattern may occur due to noise, imperfect calibration, or matching errors.

2.1.2



2.1.3

Matches with deviation > 2 pixels: 159/500

Percentage of matches with deviation > 2 pixels: 31.80%

This relatively high percentage indicates many incorrect matches, highlighting the need for geometric filtering (as addressed in Section 2.2).

2.2.1



2.2.2

Inliers: 341

Outliers: 159

Discarded matches: 159

2.2.3

Assuming erroneous matches are uniformly distributed across the image, the vertical location of a mismatched feature in the second image is equally likely to be anywhere along the image height.

The rectified stereo constraint accepts matches only if:

$$|y_{\text{left}} - y_{\text{right}}| \leq 2 \text{ pixels}$$

This corresponds to an acceptance band of 5 pixels (from -2 to +2) along the vertical axis.

Given the image height $H = 376$ pixels, the probability that a random erroneous match falls within this band is:

$$5 / 376 \approx 0.0133 \approx 1.33\%$$

Therefore, the expected rejection ratio is:

$$1 - 0.0133 \approx 0.9867 \approx 98.67\%$$

In our case, 159 matches were rejected. Assuming these correspond to erroneous matches, the expected number of erroneous matches that are wrongly accepted is:

$$159 \times (0.0133 / 0.9867) \approx 2.1$$

That is, approximately 2 erroneous matches are expected to be incorrectly accepted.

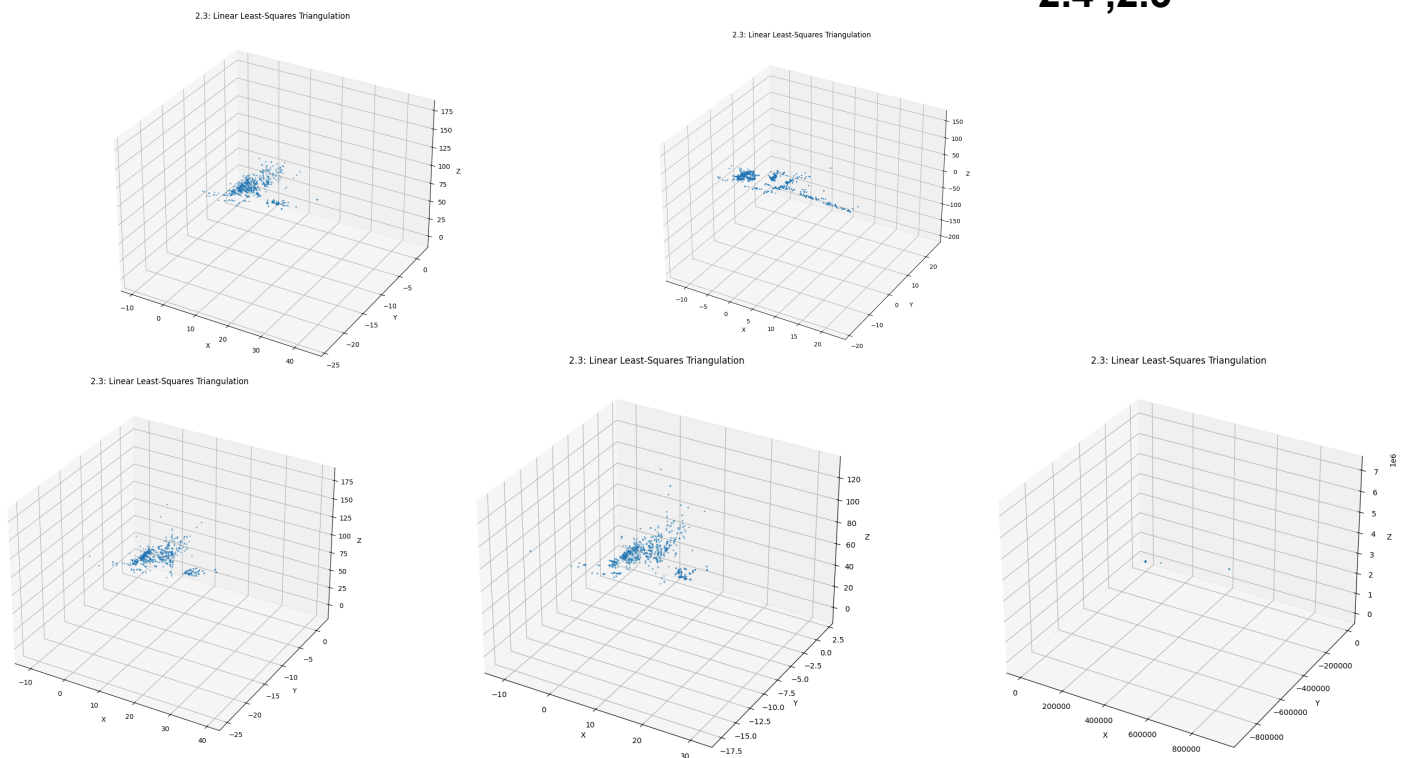
2.2.4

The assumption of uniform distribution is not realistic. As can be seen in the image(2.2.3), the matches are not uniformly spread across the image, but rather concentrated in specific regions such as building edges, trees, and textured areas.

Erroneous matches tend to occur in visually similar or repetitive structures, which often lie along similar horizontal lines. As a result, many incorrect matches still satisfy the rectified stereo constraint (small vertical deviation).

Therefore, the actual number of accepted erroneous matches is expected to be higher than the theoretical estimate, since the uniform distribution assumption underestimates the likelihood of incorrect matches falling within the allowed vertical band.

2.4 ,2.3



Frames 0–4 arranged from left to right and top to bottom.

The erroneous 3D points are most likely caused by incorrect feature matches and triangulation instability. In particular, mismatched keypoints cause the viewing rays from the two cameras to intersect poorly or become nearly parallel, leading to extremely large or unstable depth estimates.

A relevant criterion for outlier removal is the rectified stereo constraint, requiring matched points to satisfy approximately the same vertical image coordinate:

$$|Y_{\text{left}} - Y_{\text{right}}| \leq \epsilon$$

Since correct matches in rectified stereo images should lie on the same epipolar line. Another useful criterion could be reprojection error after triangulation.